

Extended Supplementary Material for “Neighbor-Corroborated Kalman Filtering for Robust Distributed Multi-Object Tracking”

This document accompanies the preprint available at <https://doi.org/10.22541/au.177490735.56661375/v1>.

I. ADAPTIVE ADVERSARIAL ATTACKS

To evaluate the robustness of KF-ALARM against informed adversaries, an adaptive attack scenario is considered. In this setting, the adversary is assumed to have knowledge of the filtering mechanism and seeks to weaken the corroboration process by targeting nodes with limited connectivity and progressively compromising their neighbors. This reduces the availability of reliable cross-node support and degrades the effectiveness of the corroboration mechanism.

Node 6, which has a low degree (3), is selected as the primary target. The attack is then progressively intensified by expanding the set of compromised nodes as follows: (i) node 6 only; (ii) nodes 6 and 7; (iii) nodes 6, 7, and 2; and (iv) nodes 6, 7, 2, and 5. In each case, deception attacks with four ghost targets are applied by injecting measurements that mimic the motion of genuine targets.

TABLE I
KF-ALARM PERFORMANCE UNDER ADAPTIVE ATTACKS WITH PROGRESSIVELY COMPROMISED NODES (WINDOW=10, CUT-OFF=100 M, AND ORDER=1). LOWER VALUES INDICATE BETTER PERFORMANCE.

Metric	Compromised nodes			
	6	6,7	6,7,2	6,7,2,5
OSPA distance (m)	30.84	35.94	60.14	60.22
OSPA cardinality	7.83	14.28	49.19	49.28
OSPA ⁽²⁾ distance (m)	46.33	52.08	65.97	66.21
OSPA ⁽²⁾ cardinality	14.41	23.46	48.44	48.56

Table I summarizes the performance of KF-ALARM at the targeted node (6) under these adaptive attacks. When only node 6 is compromised, KF-ALARM maintains relatively stable tracking performance, indicating that corroboration from benign neighbors remains effective in suppressing adversarial measurements. As additional neighboring nodes are compromised, performance degrades progressively. In particular, the cardinality errors increase significantly when three or more nodes are attacked, reflecting the reduced availability of reliable corroboration and the increased influence of coordinated ghost measurements.

Notably, the degradation is gradual rather than abrupt, demonstrating that KF-ALARM retains partial robustness even under adversarial expansion. However, as the majority of neighboring nodes become compromised, the corroboration mechanism weakens, leading to increased false tracks and higher estimation error. These results are consistent with the expected behavior of consensus-based defenses under adaptive adversarial strategies.

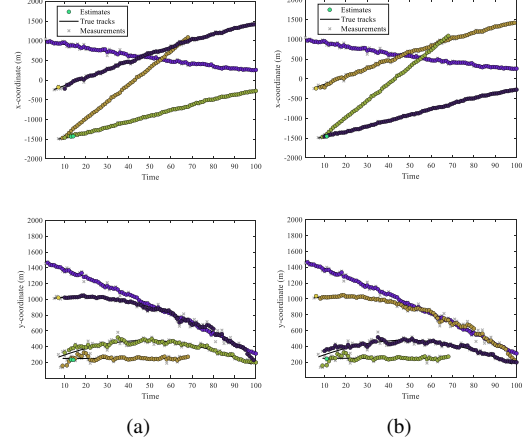


Fig. 1. Tracking performance under benign conditions: (a) KF-ALARM, and (b) the standard EKF.

II. BENIGN SCENARIO PERFORMANCE

This section reports performance under benign conditions, that is, in the absence of adversarial manipulation. The purpose is to verify that KF-ALARM preserves nominal tracking performance relative to the standard EKF baseline.

Performance is evaluated using the OSPA distance, the OSPA⁽²⁾ distance, and their associated cardinality components. These metrics quantify instantaneous multi-object error, trajectory consistency over a finite temporal window, and target-count accuracy, respectively.

Table II summarizes the average benign-case performance. KF-ALARM produces slightly larger distance errors. This behavior arises because the corroboration factor effectively inflates the measurement covariance when it is less than one, reducing the Kalman gain and yielding more conservative updates. As a result, measurements are not fully trusted even in benign conditions, leading to a small loss in estimation accuracy. Fig. 1 illustrates the tracking performance of KF-ALARM and the standard EKF under benign conditions.

TABLE II
AVERAGE TRACKING PERFORMANCE IN THE BENIGN SCENARIO ($W = 10$, $c = 100$ m, $p = 1$). LOWER VALUES INDICATE BETTER PERFORMANCE. BEST RESULTS ARE SHOWN IN BOLD.

Metric	KF-ALARM	Standard
OSPA Distance Mean (m)	25.19	24.44
OSPA Cardinality Mean	6.98	5.78
OSPA ⁽²⁾ Distance Mean (m)	34.40	33.67
OSPA ⁽²⁾ Cardinality Mean	7	5.45

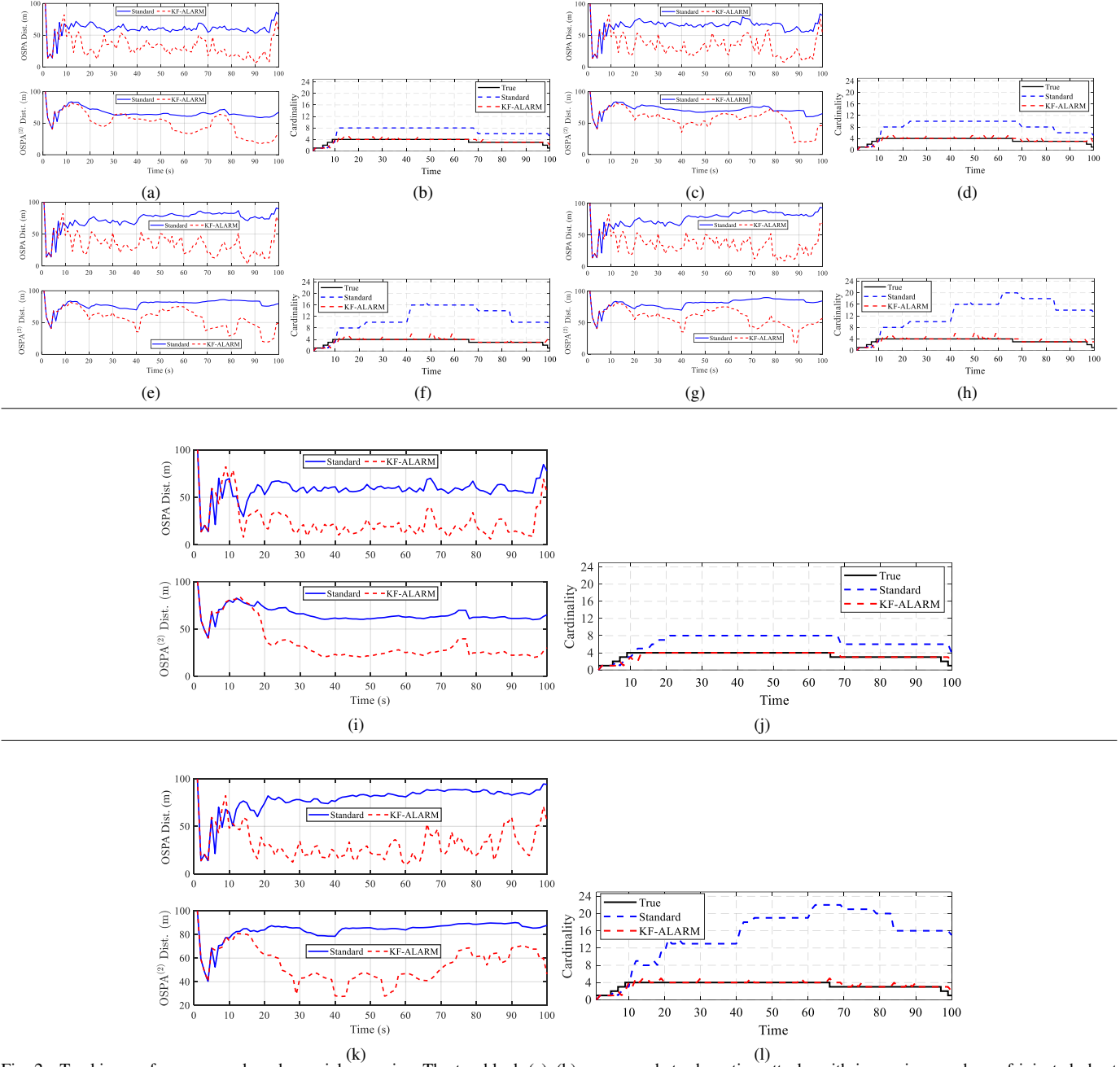


Fig. 2. Tracking performance under adversarial scenarios. The top block (a)–(h) corresponds to deception attacks with increasing numbers of injected ghost targets: (a)–(b) 4 ghosts, (c)–(d) 6 ghosts, (e)–(f) 12 ghosts, and (g)–(h) 16 ghosts. Within each pair, the left panel shows temporal OSPA and $OSPA^{(2)}$ distances, and the right panel shows the estimated and true target cardinalities. The middle block (i)–(j) shows the replay attack, and the bottom block (k)–(l) shows the flooding attack; in both cases, the left panel reports OSPA and $OSPA^{(2)}$, and the right panel reports cardinality.

III. DETAILED RESULTS UNDER ADVERSARIAL ATTACKS

This section provides additional results under deception, replay, and flooding attacks, summarized in Fig. 2. The figure reports temporal OSPA and $OSPA^{(2)}$ distances together with the estimated target cardinality, enabling a detailed assessment of tracking accuracy and cardinality stability over time.

Across all attack scenarios, the standard EKF exhibits progressively larger tracking errors and noticeable cardinality inflation as the number or severity of adversarial measurements increases. In contrast, KF-ALARM maintains lower errors and more accurate cardinality estimates, demonstrating improved robustness to adversarial measurement corruption.